Fisher Information Geometry and Conditional Implementation Costs of Self-Referential Calibration

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Abstract

Papers I and II of the T-DOME series [1, 2] supply conditional resource and reference-frame models for persistent agents. They do not prove that all persistent agents require non-Markovian memory or spontaneous Clifford gauge-symmetry breaking. Paper II's linear-drift visibility crossover is likewise a declared coordinate model, not a universal logarithmic "Delusion Trap" timescale.

This paper studies **self-referential calibration** under explicit residual-likelihood, prior, drift, convexity, and hardware assumptions. The agent uses the score statistics of a supplied prediction-residual model to estimate frame mismatch. The principal conditional results are:

1. **Gaussian Fisher Signal** (Proposition 14): in the declared even Gaussian residual model, per-sample local Fisher information grows quadratically and accumulated information grows cubically. Detection precedes the visibility crossover if and only if the threshold condition in Corollary 16 holds.
2. **Self-Referential Cramér–Rao Bound** (Theorem 18): the van Trees lower bound applies under the stated regular likelihood, conditional-independence, compact-support, and prior assumptions, with data and prior Fisher information entering separately.
3. **Conditional Implementation Cost** (Proposition 28): a separable implementation with declared irreversible sensor and computational resets and linear-response actuation obeys $\dot{W}_{\mathrm{loop}} \geq k_B T \ln 2\,(R_s + R_c) + \mathcal{L}^2/\tau^2_{\mathrm{recalib}}$, a bound for that implementation rather than a universal cost of sensing, Fisher evaluation, or self-reference.

The calibration loop has a local tracking bound only under geodesic strong convexity and bounded drift (Theorem 23). The four supplied layers in Definition 27 define the running model architecture; they are not proved necessary, sufficient, minimal, or complete for persistent agency. "Reflexivity" here denotes one implementation of second-order control, with no claim of architectural uniqueness or phenomenal consciousness.

1 Introduction

1.1 Context: The Delusion Trap

Revised Paper II [2] studies a supplied coordinate projection under a raw-channel budget. It does not prove that persistence forces Clifford gauge symmetry breaking. Its drift model uses a linearly changing mismatch $\theta(t) = \theta_0 + \varepsilon t$ and identifies the geometric visibility crossover

$$t_{\text{vis}} = \frac{\pi/4 - \theta_0}{\varepsilon}. \tag{1}$$

That crossover does not by itself imply gradient reversal or undetectability. The present paper asks a narrower question: for a specified residual likelihood, can score statistics detect mismatch, and under what conditions can a natural-gradient controller track a moving optimum?

1.2 Position within Papers I–III

This paper is the third and final of the T-DOME framework, closing the three-paper sequence.

Framework	**Question**	**Result**	**Status**
HAFF [17, 18]	How does geometry emerge?	Conditional algebra/dictionary models	Revised
Q-RAIF [20, 21]	What algebra must an observer have?	Clifford representations as model inputs	Revised
T-DOME I [1]	What can inter-epoch correlations fund?	Restricted baseline; correlation accounting	Conditional
T-DOME II [2]	When can a model select a frame?	Conditional coordinate selection	Conditional
T-DOME III (this work)	How does self-calibration arise?	Fisher self-referential bound; thermodynamic cost of reflexivity	**This paper**

The three T-DOME papers form a conditional modelling sequence:

1. **Paper I:** Initial correlations can enter an ELV-type thermodynamic balance; a restricted autonomous baseline is proved.

2. **Paper II:** A raw-coordinate archive motivates a supplied compression model and a linear visibility-crossover example.

3. **Paper III (this work):** Uncorrected bias diverges from a changing environment. A self-referential calibration loop—monitoring the Fisher information of one's own prediction stream—resolves mismatch under explicit likelihood and convexity assumptions. A conditional hardware model assigns reset and actuation costs.

The links are motivation and model interfaces, not an irreversible theorem chain. None of the three papers proves that this architecture is necessary for persistence.

1.3 The Information-Geometric Insight

The key observation that resolves the Delusion Trap is subtle: *while the agent cannot observe V_{bg} directly, it can observe the statistical properties of its own prediction residuals in V_{fg}.*

The prediction residual $e(t) := \mathcal{S}_{\text{vis}}(t) - \mathcal{S}_{\text{vis}}^{(\text{pred})}(t)$ lies in V_{fg} by construction. In the declared even Gaussian model, a record cannot identify the sign of mismatch. Its *distribution*—the probability law $p(e \,|\, \sigma)$, parametrised by the gauge-fixing parameter σ—does depend on σ, because the projection $\Pi_{\mathcal{F}}(\sigma)$ determines which environmental correlations are captured and which are discarded.

When the frame σ drifts away from the optimal σ^*, the residual distribution shifts. The *Fisher information metric*

$$g_{ij}(\sigma) = \mathbb{E}_\sigma\left[\frac{\partial \log p(e \,|\, \sigma)}{\partial \sigma^i}\,\frac{\partial \log p(e \,|\, \sigma)}{\partial \sigma^j}\right] \tag{2}$$

measures the sensitivity of this distribution to changes in σ. Whether g_{ij} changes with staleness is likelihood-dependent. The Gaussian model below provides one explicit case.

This is the mathematical realisation of the "second-order operation" demanded by Paper II, Section 7.5: the agent does not need to see the truth (the full $Cl(V, q)$), but only the *rate of change of its own prediction error* as a function of its frame parameters. Fisher information is one local sensitivity measure, not the unique possible detector.

1.4 Relation to Architectural Incompleteness

Revised HAFF Paper G [19] supplies philosophy-of-science motivation rather than a fixed-point no-go theorem. In that motivational sense, the self-referential calibration loop cannot *eliminate* architectural incompleteness, but it can *track* the consequences of incompleteness in real time. The Lyapunov function $V(\sigma)$ monitors the distance between the agent's frame and the optimal frame without requiring access to the "complete" description—it operates entirely within the agent's own predictive statistics.

1.5 Scope and Disclaimers

1. *Reflexivity* refers throughout to second-order control: the ability of a system to monitor and adjust its own monitoring process. It carries *no* implication of phenomenal consciousness, subjective experience, or qualia.

2. The self-referential calibration loop does not *eliminate* the ego's bias; it tracks and compensates for drift in the bias. The four bias terms of Paper II persist in the calibrated phase.

3. The thermodynamic cost statement is conditional on irreversible reset, separable hardware, and a linear-response actuation model.

4. The framework applies to systems satisfying (C1)–(C5) (Section 2.6). It is not a universal theory of agency.

Related work. The Fisher information metric on statistical manifolds was introduced by Rao [3] and shown to be unique by Čencov [7]. The natural gradient and information geometry were developed by Amari [6, 5]. Thermodynamic length and optimal finite-time transformations were established by Crooks [8] and Sivak–Crooks [9]. The connection between Fisher information and entropy production was formalised by Ito [12] and Barato–Seifert [11]. Second-order cybernetics originates with Ashby [13] and von Foerster [14]. Adaptive control and self-tuning regulators are treated in [15]. The Bayesian Cramér–Rao bound (van Trees inequality) is from [10].

Summary of contributions. This paper separates three levels of result:

1. **Drift Detectability** (Proposition 14): an explicit Gaussian residual likelihood has quadratic per-sample and cubic accumulated local Fisher signals; early detection is a threshold condition.

2. **Self-Referential Cramér–Rao Bound** (Theorem 18): drift-estimation precision is bounded by the sum of data Fisher information and ego rigidity.

3. **Thermodynamic Cost** (Proposition 28): a separable implementation with irreversible resets and finite-time actuation obeys a conditional additive bound.

Result-status ledger. The original drift-detectability theorem `TD-III-T3.3` is retracted; Proposition 14 is an FD/I model calculation and Corollary 16 is conditional on a declared threshold. Ego rigidity `TD-III-ego` remains a supplied prior model. The tracking theorem is conditional on local geodesic strong convexity. The universal additive-cost claim `TD-III-T6.2` is downgraded to Proposition 28, and the inherited persistence budget is withdrawn.

2 Mathematical Preliminaries

2.1 Inherited Framework from Papers I and II

We briefly recall the key objects; the reader is referred to Papers I and II for full definitions and proofs.

From Paper I [1].

- **Survival functional.** $\mathcal{S}[\Lambda, \tau] := \Delta F - W[0, \tau]$ (Paper I, Eq. (9)).
- **Restricted autonomous baseline.** $\mathcal{S}[\Lambda^{\mathrm{D}}, \tau] \leq 0$ for a fixed-Hamiltonian, no-control Davies semigroup under its stated initial conditions. It is not a ceiling for all Markovian agents.

- **Memory kernel.** $\mathcal{K}(t,s)$: the non-Markovian superoperator encoding system–environment correlations.
- **Entropy rate.** $h_\mu := \lim_{T\to\infty} T^{-1} H(X_{0:T})$ (bits per unit time per algebraic component).
- **Predictive information.** $I_{\text{pred}} := I(\overleftarrow{X};\, \overrightarrow{X})$.

From Paper II [2].

- **Internal algebra.** $\mathcal{O}_{\text{int}} = Cl(V,q)$, $D = \dim Cl(V,q) = 2^n$, gauge group $G = \text{Aut}(Cl(V,q))$.
- **Gauge bundle.** $\pi : P \to M$, structure group G; a section $\sigma : M \to P$ is a reference frame.
- **Ego.** $\mathfrak{E} := (\mathcal{F}^*, V_{\text{fg}}^*)$ denotes a chosen optimizer in a supplied encoder class, with $k^* = k_{\text{opt}} \le k_{\max}$. The budget capacity is not a universal optimizer formula.
- **Projected kernel.** $\mathcal{K}_{\mathcal{F}}(t,s) = \Pi_{\mathcal{F}}\, \mathcal{K}(t,s)\, \Pi_{\mathcal{F}}$.
- **Survival decomposition.** $\mathcal{S} = \mathcal{S}_{\text{vis}}(\mathcal{F}) + \mathcal{S}_{\text{hid}}(\mathcal{F})$.
- **Four bias terms.** $\mathcal{B}_{\text{select}}$, $\mathcal{B}_{\text{frame}}$, $\mathcal{B}_{\text{center}}$, $\mathcal{B}_{\text{inc}}$ (revised Paper II, four-diagnostics definition). These are model diagnostics, not generic consequences of gauge fixing.
- **Visibility crossover.** $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$ in the supplied linear drift geometry. Gradient reversal requires an additional controller condition.
- **Information objects.** Mutual information is evaluated on explicitly induced record processes; algebra symbols are not themselves random variables.

2.2 Fisher Information Metric

Definition 1 (Fisher information matrix)**.** *Let* $\{p(x\,|\,\theta) : \theta \in \Theta \subset \mathbb{R}^d\}$ *be a parametric family of probability densities satisfying standard regularity conditions (interchange of differentiation and integration). The* Fisher information matrix *is*

$$g_{ij}(\theta) := \mathbb{E}_\theta\left[\frac{\partial \log p(x\,|\,\theta)}{\partial\theta^i}\,\frac{\partial \log p(x\,|\,\theta)}{\partial\theta^j}\right] = -\mathbb{E}_\theta\left[\frac{\partial^2 \log p(x\,|\,\theta)}{\partial\theta^i\,\partial\theta^j}\right]. \tag{3}$$

The pair $(\Theta,\, g)$ *is a Riemannian manifold called the* statistical manifold.

Remark 2 (Uniqueness)**.** *By Čencov's theorem [7], the Fisher–Rao metric* g^{FR} *is, up to a positive scalar multiple, the unique Riemannian metric on the space of probability distributions that is invariant under all Markov morphisms (sufficient-statistic embeddings). Within the hypotheses and category used by Čencov's theorem, this makes the Fisher metric a canonical local distinguishability metric. It does not make a Fisher-based detector physically mandatory, nor does it choose a likelihood or control objective for the agent.*

Proposition 3 (Cramér–Rao bound)**.** *For any unbiased estimator $\hat{\theta}$ of θ based on n independent observations:*

$$\mathrm{Cov}(\hat{\theta}) \;\succeq\; \frac{1}{n}\,\big[g(\theta)\big]^{-1} \tag{4}$$

in the Löwner order. The scalar case reads $\mathrm{Var}(\hat{\theta}) \geq 1/\big(n\,g(\theta)\big)$.

Remark 4 (Effective independence)**.** *Throughout this paper, references to "independent observations" in the context of continuous-time residual streams should be read as* effective independence *after thinning by the environmental decorrelation time τ_E, yielding an effective sample size $n_{\mathrm{eff}} \approx T/\tau_E$. In particular, the sample count n in* (4) *becomes n_{eff} in the self-referential setting of Section 4.2.*

Remark 5 (Fisher metric and KL divergence)**.** *The Fisher metric arises as the Hessian of the Kullback–Leibler divergence [16]:*

$$D_{\mathrm{KL}}\big(p_\theta \,\|\, p_{\theta+d\theta}\big) = \tfrac{1}{2}\, g_{ij}(\theta)\, d\theta^i\, d\theta^j + O\big(|d\theta|^3\big)\,. \tag{5}$$

This identifies the Fisher metric as the infinitesimal measure of statistical distinguishability.

2.3 Information Geometry

Following Amari [4, 5], the statistical manifold $(\Theta,\, g)$ carries additional geometric structure beyond the Riemannian metric.

α-connections. For each $\alpha \in [-1,\, 1]$, Amari defines an affine connection $\nabla^{(\alpha)}$ on Θ. The cases $\alpha = 1$ (exponential connection, $\nabla^{(e)}$) and $\alpha = -1$ (mixture connection, $\nabla^{(m)}$) are *dual* with respect to g: $\partial_k\, g(X, Y) = g(\nabla^{(e)}_k X,\, Y) + g(X,\, \nabla^{(m)}_k Y)$. For exponential families, $\nabla^{(e)}$ is flat in natural parameters and $\nabla^{(m)}$ is flat in expectation parameters—the *dually flat structure*. The case $\alpha = 0$ recovers the Levi-Civita connection of the Fisher metric.

Natural gradient. Standard gradient descent in parameter space ignores the curvature of the statistical manifold. The *natural gradient* [6]

$$\dot{\theta} = -\eta\, g^{-1}(\theta)\, \nabla_\theta L(\theta), \tag{6}$$

where $\eta > 0$ is the learning rate and $L(\theta)$ is a loss function, provides the steepest descent direction in the Fisher metric. It is reparametrisation-invariant. Statistical efficiency requires additional estimator and asymptotic assumptions and is not implied by the update formula alone.

Pythagorean theorem. In a dually flat space, the KL divergence satisfies a generalised Pythagorean relation: $D_{\mathrm{KL}}(p \,\|\, r) = D_{\mathrm{KL}}(p \,\|\, q) + D_{\mathrm{KL}}(q \,\|\, r)$ when q is the m-projection of p onto a submanifold containing r. This decomposition will be applied to separate the foreground-recoverable and background-irrecoverable components of drift.

2.4 Thermodynamic Length

Definition 6 (Thermodynamic length)**.** *Let $\lambda(t)$ for $t \in [0, \tau]$ be a path through control parameter space, and let $\zeta_{ij}(\lambda)$ be the* friction tensor *(the time-integrated equilibrium force–force correlation function at λ). The* thermodynamic length *of the path [8] is*

$$\mathcal{L} := \int_0^\tau \sqrt{\zeta_{ij}(\lambda)\, \dot{\lambda}^i\, \dot{\lambda}^j}\; dt. \tag{7}$$

Proposition 7 (Sivak–Crooks bound)**.** *The excess (dissipated) work during a finite-time transformation of duration τ satisfies [9]*

$$W_{\text{ex}} \;\geq\; \frac{\mathcal{L}^2}{\tau}. \tag{8}$$

The minimum is achieved by the geodesic of the friction tensor ζ. In the linear-response regime, the friction tensor is related to the Fisher metric of the equilibrium distribution at λ by $\zeta_{ij}(\lambda) \sim \tau_{\text{relax}}\, g_{ij}^{\text{Fisher}}(\lambda)$, where τ_{relax} is the relaxation time.

2.5 Second-Order Cybernetics

Von Foerster [14] distinguished two levels of control:

- **First-order cybernetics**: feedback control of observed systems. The controller adjusts its actions based on the output of a sensor. Paper II's ego is a first-order structure: it processes environmental data within a fixed frame.
- **Second-order cybernetics**: feedback control of the *observing* system itself. The controller adjusts the *sensor*—or equivalently, the reference frame within which the sensor operates. This is what Paper III provides.

Ashby's Law of Requisite Variety [13] provides a lower bound on the complexity of the meta-controller:

$$\dim(\text{meta-controller state space}) \;\geq\; \dim(\text{environmental drift subspace})\,. \tag{9}$$

The meta-observer must have at least as many adjustable parameters as there are independent modes of environmental drift.

In adaptive control theory [15], the analogous result is the *persistent excitation* condition: parameter estimates converge if and only if the input signal is "rich enough" to excite all modes of the system. In our framework, persistent excitation corresponds to $h_\mu > 0$—the environment must continue to generate novelty for the self-calibration loop to function.

Remark 8 (Operational content)**.** *The second-order cybernetic structure in this paper is* not *a philosophical metaphor. It has concrete operational content: the natural gradient update* (6) *is a specific algorithm that takes as input the Fisher information of the residual stream and produces as output an update to the frame parameter σ. An implementation must supply a differentiable residual likelihood, evaluate or approximate its score, and realize the update law. Merely accumulating an unlabelled residual second moment is not generally a Fisher computation.*

2.6 Standing Assumptions

Definition 9 (Standing Assumptions)**.** *Throughout this paper, the following conditions are assumed:*

(C1) ***Supplied representation model.*** *A finite-dimensional agent model, a foreground projection, and a control algebra are specified. A Clifford representation and any external map ϕ are model inputs, not conclusions inherited from revised Q-RAIF or T-DOME II.*

(C2) ***Environmental drift.*** *The instantaneous optimal frame $\mathcal{F}^*(t)$ rotates continuously in G at bounded speed. In the scalar worked model the mismatch is $\theta(t) = \theta_0 + \varepsilon t$ up to the first visibility crossover, as in revised Paper II.*

(C3) ***Finite meta-observer budget.*** *The self-calibration loop has a computational budget $\mathcal{C}_{\text{meta}} < \infty$ (bits per unit time), distinct from the ego's processing budget $\mathcal{C}_{\text{budget}}$.*

(C4) ***Regularity.*** *The agent's predictive family $\{p(e \,|\, \sigma) : \sigma \in G/H\}$ satisfies standard Fisher information regularity: finite Fisher matrix and interchange of differentiation and integration. Any scalar sensitivity or uniform lower bound used later must be assumed separately.*

(C5) ***Persistent excitation.*** *The time-averaged Fisher matrix obeys the explicit lower bound in Definition 24. A positive entropy rate alone is not sufficient for this condition.*

3 The Drift Detection Problem

3.1 Identifiability in an Even Residual Model

Proposition 10 (Sign non-identifiability in an even residual model)**.** *Suppose the scalar residual model is $e \mid \theta \sim \mathcal{N}(a\theta^2, \sigma_e^2)$ with $a \neq 0$ and $\sigma_e > 0$. Then $p(e \mid \theta) = p(e \mid -\theta)$ for every sample and every record length. Residual data alone therefore cannot identify the sign of the frame mismatch without an orientation-breaking input.*

Proof. The likelihood depends on θ only through θ^2; hence the joint likelihood of any record is invariant under $\theta \mapsto -\theta$. No statistic of that record can distinguish the two parameter values. □

This proposition is a property of the declared even likelihood. It does not prove that every instantaneous or "first-order" controller fails: directional probes, asymmetric residual models, and exogenous orientation data can remove the ambiguity.

3.2 The Agent's Statistical Manifold

The agent's prediction-residual stream $\{e(t)\}_{t\geq 0}$ defines a stochastic process whose distribution depends on the gauge-fixing parameter σ. We model this dependence as a parametric family.

Definition 11 (Predictive family)**.** *The* predictive family *of the agent is the set*

$$\mathcal{P} := \big\{p(e \,|\, \sigma) : \sigma \in \mathcal{M}_G\big\}, \tag{10}$$

where $\mathcal{M}_G := G/H$ is the space of gauge-fixing orbits (H is the stabiliser of the foreground subspace), e denotes the prediction-residual time series over a window of length T, and $p(e \,|\, \sigma)$ is the likelihood of the observed residuals given the gauge parameter σ.

The key insight is that $p(e \,|\, \sigma)$ depends on σ even though $e(t) \in V_{\text{fg}}$, because the projection $\Pi_{\mathcal{F}}(\sigma)$ determines which environmental correlations are captured. When σ drifts from the optimal σ^*:

- The variance of the residuals increases (the discarded background components contribute unmodelled noise).
- The temporal correlations of the residuals change (the projected kernel $\mathcal{K}_{\mathcal{F}}$ no longer captures the dominant environmental modes).
- Higher-order statistics (kurtosis, spectral shape) shift systematically.

These distributional changes are invisible to the raw error $e(t)$ but detectable by the Fisher metric of $\mathcal{P}$.

3.3 Self-Referential Fisher Information

Definition 12 (Self-referential Fisher information)**.** *For one residual with likelihood $p(e \mid \sigma)$, define the* per-sample Fisher information

$$\mathcal{I}_1(\sigma) := \mathbb{E}_\sigma\big[(\partial_\sigma \log p(e \mid \sigma))^2\big]\,. \tag{11}$$

For conditionally independent residuals observed at times t_m, the accumulated local information *is $J_T := \sum_{t_m \le T} \mathcal{I}_1(\sigma(t_m))$. The Fisher metric g, the local quadratic form $g_{ij}\delta\sigma^i\delta\sigma^j$, and J_T are distinct objects. Fisher information is measured in inverse-squared parameter units, not bits.*

Remark 13 (What the agent "measures")**.** *Computing $\mathcal{I}_1(\sigma)$ does not require access to V_{bg} or to the "true" environment. It requires only: (i) the agent's own prediction residuals $\{e(t)\}$ (which lie in V_{fg}), and (ii) the ability to evaluate the score function $\partial_\sigma \log p(e \,|\, \sigma)$—the sensitivity of its own predictive model to frame perturbations. This is a computation entirely within the agent's internal algebra, using only quantities already available from the ego's processing pipeline.*

Proposition 14 (Fisher signal in a Gaussian residual model)**.** *Let residuals sampled at rate r be conditionally independent with*

$$e_m \mid \theta_m \sim \mathcal{N}(a\theta_m^2, \sigma_e^2), \qquad \theta(t) = \theta_0 + \varepsilon t, \tag{12}$$

where $a \neq 0$, $\sigma_e > 0$, and $\varepsilon > 0$. The per-sample Fisher information with respect to the current mismatch is

$$\mathcal{I}_1(\theta) = \frac{4a^2\theta^2}{\sigma_e^2}. \tag{13}$$

Consequently the expected accumulated local information over a window of length T is

$$J_T \simeq \frac{4a^2 r}{3\sigma_e^2 \varepsilon} \left[(\theta_0 + \varepsilon T)^3 - \theta_0^3 \right]. \tag{14}$$

For a freshly calibrated frame ($\theta_0 = 0$), the per-sample signal is quadratic in time and $J_T = 4a^2 r \varepsilon^2 T^3/(3\sigma_e^2)$.

Proof. For a normal family with fixed variance and mean $\mu(\theta) = a\theta^2$, the score is $(e - \mu)\mu'(\theta)/\sigma_e^2$. Taking its squared expectation gives $(\mu')^2/\sigma_e^2$, which is (13). Summing over samples and using the sampling rate gives $J_T \simeq r \int_0^T \mathcal{I}_1(\theta_0 + \varepsilon t) dt$, which evaluates to (14). □

Remark 15 (Scope of the sensitivity claim)**.** *Full-rank Fisher information for σ does not imply that a scalar mismatch map $\theta(\sigma)$ has a Jacobian bounded away from zero. Proposition 14 therefore works directly in the identified scalar mismatch coordinate. A multi-parameter extension must supply or derive the relevant sensitivity map.*

Corollary 16 (Conditional detection before visibility crossover)**.** *In the Gaussian residual model, let $J_{\min} > 0$ be a declared detection threshold. For a freshly calibrated frame,*

$$T_{\text{det}} = \left(\frac{3\sigma_e^2 J_{\min}}{4a^2 r \varepsilon^2} \right)^{1/3}. \tag{15}$$

Detection precedes Paper II's visibility crossover if and only if $T_{\text{det}} < t_{\text{vis}}$, where for the same linear drift $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$. Equivalently, for general θ_0 the condition is $J_{t_{\text{vis}}} > J_{\min}$. No unconditional ordering follows from regularity alone.

Proof. Set the fresh-calibration expression for J_T in Proposition 14 equal to $J_{\min}$ and solve for T. Comparing that value with the independently defined visibility time gives the stated condition. □

4 The Self-Referential Bound

4.1 The Bayesian Framework

The agent's ego structure (Paper II) provides a *prior belief* about the correct gauge parameter: the current frame σ_0 is the ego's "preferred" value. We encode this as a prior distribution $\pi_{\text{ego}}(\sigma)$, concentrated around σ_0.

Definition 17 (Ego rigidity)**.** *The* ego rigidity *is the prior Fisher information*

$$\mathcal{I}_{\text{ego}} := \int_{\mathcal{M}_G} \left(\frac{\partial \log \pi_{\text{ego}}(\sigma)}{\partial \sigma} \right)^2 \pi_{\text{ego}}(\sigma)\, d\sigma. \tag{16}$$

High $\mathcal{I}_{\text{ego}}$ corresponds to a sharply peaked prior (rigid ego); low $\mathcal{I}_{\text{ego}}$ to a diffuse prior (flexible ego). This prior and its Fisher information are supplied model data. The diagnostics of Paper II do not determine a prior without an additional statistical construction.

4.2 The Self-Referential Cramér–Rao Bound

Theorem 18 (Self-Referential Cramér–Rao Bound)**.** *Let the scalar drift parameter $\delta\sigma$ lie in a compact interval $[a, b]$. Assume that the residual likelihood is regular, that the effective samples are conditionally independent, and that the prior π_{ego} is C^1, has finite prior Fisher information, and vanishes at a and b. Let $\delta\hat{\sigma}$ be any estimator based only on a residual record of duration T. Define the* effective sample size $n_{\text{eff}} := T/\tau_E$*, where τ_E is the decorrelation time of the residual process $\{e(t)\}$ (the time beyond which consecutive residuals carry approximately independent information about σ). If the effective samples share per-sample information $\mathcal{I}_1(\delta\sigma)$, the van Trees inequality [10] gives*

$$\mathbb{E}\Big[\big|\delta\hat{\sigma} - \delta\sigma\big|^2\Big] \;\geq\; \frac{1}{n_{\text{eff}}\,\mathbb{E}_\pi[\mathcal{I}_1(\delta\sigma)] + \mathcal{I}_{\text{ego}}}. \tag{17}$$

Proof. This is a direct application of the van Trees (Bayesian Cramér–Rao) inequality [10]. The total information about the drift parameter $\delta\sigma$ consists of two contributions:

- $n_{\text{eff}}\,\mathbb{E}_\pi[\mathcal{I}_1]$: the data Fisher information. In continuous time, the residual process is correlated with decorrelation time τ_E set by the bath memory kernel. Over a window of duration T, the process yields $n_{\text{eff}} \approx T/\tau_E$ effectively independent samples, each carrying Fisher information about $\delta\sigma$.
- $\mathcal{I}_{\text{ego}}$: the prior Fisher information from the ego's preference for σ_0 (Definition 17).

The boundary and regularity assumptions remove the integration-by-parts boundary term in the standard van Trees proof. Fisher information is not measured in bits; the denominator has inverse-squared parameter units. □

Remark 19 (The ego as help and hindrance)**.** *The ego rigidity $\mathcal{I}_{\text{ego}}$ acts as both help and hindrance:*

- ***Help****: when the ego is well-aligned ($\sigma_0 \approx \sigma^*$), the prior tightens the bound, reducing estimation variance.*
- ***Hindrance****: when the ego is misaligned ($|\sigma_0 - \sigma^*|$ large), the prior pulls the estimate toward the wrong value, creating a* confirmation bias *that resists recalibration.*

For a conjugate scalar Gaussian location model only, the posterior mean has the precision-weighted form below, where $\bar{\mathcal{I}}_1$ is its constant per-sample data precision:

$$\hat{\sigma}_{\text{opt}} = \frac{n_{\text{eff}}\,\bar{\mathcal{I}}_1\,\hat{\sigma}_{\text{MLE}} + \mathcal{I}_{\text{ego}}\,\sigma_0}{n_{\text{eff}}\,\bar{\mathcal{I}}_1 + \mathcal{I}_{\text{ego}}}, \tag{18}$$

a weighted average of the maximum-likelihood estimate $\hat{\sigma}_{\text{MLE}}$ and the ego's prior belief σ_0, with weights proportional to their respective Fisher informations. This formula is not implied by van Trees for a general likelihood or prior. In the conjugate Gaussian case, as $n_{\text{eff}}\,\bar{\mathcal{I}}_1 \gg \mathcal{I}_{\text{ego}}$ (enough data to overwhelm the ego), the estimator converges to the MLE.

4.3 The Rigidity-Sensitivity Trade-off

Proposition 20 (Optimal ego rigidity)**.** *Let the total expected loss be* $\mathcal{L}_{\text{total}}(\mathcal{I}_{\text{ego}}) = \mathcal{L}_{\text{estimation}} + \lambda\,\mathcal{L}_{\text{calibration}}$*, where* $\mathcal{L}_{\text{estimation}}$ *is the mean-squared drift-estimation error (bounded by* (17)*) and* $\mathcal{L}_{\text{calibration}}$ *is the cost of adjusting the frame (proportional to the frame rotation distance, hence larger when the ego is rigid and must be overcome). If the admissible rigidity interval is nonempty and compact and the two supplied loss terms are lower semicontinuous, an optimal rigidity* $\mathcal{I}^*_{\text{ego}}$ *exists. Uniqueness or an interior optimum additionally requires a model-specific strict-convexity or monotonicity condition.*

Too rigid *(*$\mathcal{I}_{\text{ego}} \gg n_{\text{eff}}\,\bar{\mathcal{I}}_1$*): the ego overwhelms the data; the agent is blind to drift.* Too soft *(*$\mathcal{I}_{\text{ego}} \ll n_{\text{eff}}\,\bar{\mathcal{I}}_1$*): the agent overreacts to noise; calibration cost is high. The optimum balances sensitivity against stability.*

Proof. A lower-semicontinuous real function attains its minimum on a nonempty compact set. The existence conclusion uses only that fact; the qualitative "too rigid/too soft" interpretation is not a convexity proof. □

5 The Calibration Loop

5.1 The Natural Gradient Update Law

The meta-observer updates the gauge parameter σ following the natural gradient on the statistical manifold $(\mathcal{M}_G,\, g)$:

$$\dot{\sigma} = -\eta\, g^{-1}(\sigma)\, \nabla_\sigma L_{\text{frame}}(\sigma), \tag{19}$$

where $\eta > 0$ is the adaptation rate and the *frame loss* is

$$L_{\text{frame}}(\sigma) := \mathbb{E}_\sigma[-\mathcal{S}_{\text{vis}}(\sigma)]\,. \tag{20}$$

The frame loss is minimised at the optimal gauge σ^* that maximises visible survival. The Fisher metric enters through the inverse g^{-1} in the natural gradient, not as a penalty term: it defines the *geometry* of the update, ensuring reparametrisation invariance.

Remark 21 (Reparametrisation invariance)**.** *The natural gradient* (19) *is invariant under reparametrisation of the gauge manifold* $\mathcal{M}_G$*: the update direction does not depend on the choice of coordinates for* σ*. This is essential because the gauge manifold inherits its geometry from the Clifford algebra, and no canonical coordinate system is preferred.*

5.2 Lyapunov Stability of the Loop

Drift velocity. Let $\sigma^*(t)$ denote the instantaneous optimal gauge parameter, defined as a minimiser of L_{frame} at time t in Eq. (20). Define the *drift velocity* $\dot{\sigma}^* := d\sigma^*/dt$, measured with respect to the Fisher metric g; its norm $\|\dot{\sigma}^*\|_g := \sqrt{g_{ij}\,\dot{\sigma}^{*i}\,\dot{\sigma}^{*j}}$ is the instantaneous rate at which the environment's optimal frame rotates on the gauge manifold.

Definition 22 (Lyapunov monitoring function)**.** *The* Lyapunov monitoring function *is the squared geodesic distance on the statistical manifold from the current frame to the instantaneous optimal frame:*

$$V(\sigma) := d_{\mathrm{geo}}(\sigma,\, \sigma^*(t))^2\,, \tag{21}$$

where d_{geo} is the geodesic distance in the Fisher metric g.

Theorem 23 (Loop Tracking Bound)**.** *Assume the natural-gradient trajectory remains in a geodesically convex normal neighbourhood of $\sigma^*(t)$ and that the frame loss is geodesically m_L-strongly convex there, with* $\operatorname{Hess} L_{\mathrm{frame}} \succeq m_L g$ *for some $m_L > 0$. Then the update* (19) *satisfies*

$$\frac{dV}{dt} \;\le\; -2\eta\, m_L\, V \;+\; 2\sqrt{V}\,\left\|\dot\sigma^*\right\|_g, \tag{22}$$

where $\|\dot\sigma^*\|_g$ is the instantaneous drift speed of the optimal frame. Consequently:

(a) ***Tracking.*** *Whenever $\sqrt{V} > \|\dot\sigma^*\|_g/(\eta\, m_L)$, we have $dV/dt < 0$: the loop actively reduces the mismatch.*

(b) ***Tracking neighbourhood.*** *The mismatch converges to a neighbourhood of the set of stationary points of L_{frame}. This neighbourhood has size*

$$V_\infty \;:=\; \frac{\left\|\dot\sigma^*\right\|_g^2}{(\eta\, m_L)^2}. \tag{23}$$

For bounded drift ($\|\dot\sigma^*\|_g \le \Lambda_{\max}$), the mismatch is bounded: $\limsup_{t\to\infty} V(t) \le \Lambda_{\max}^2/(\eta m_L)^2$.

(c) ***Static limit.*** *When $\sigma^* = \mathrm{const}$ ($\dot\sigma^* = 0$), the bound reduces to $dV/dt \le -2\eta m_L\, V$, giving exponential convergence $V(t) \le V(0)\, e^{-2\eta m_L\, t}$.*

Proof. Since $V(\sigma) = d_{\mathrm{geo}}(\sigma,\, \sigma^*(t))^2$ and $\sigma^*(t)$ is time-varying, the total derivative has two contributions:

$$\frac{dV}{dt} = \underbrace{\frac{\partial V}{\partial \sigma}\cdot\dot\sigma}_{\text{control}} \;+\; \underbrace{\frac{\partial V}{\partial \sigma^*}\cdot\dot\sigma^*}_{\text{drift}}.$$

Control term. In normal coordinates centred at σ^*, let $\delta\sigma := \sigma - \sigma^*$. The control contribution is $2\, g(\delta\sigma,\, \dot\sigma) \;=\; 2\, g(\delta\sigma,\, -\eta\, g^{-1}\nabla L_{\mathrm{frame}}) \;=\; -2\eta\, \langle \delta\sigma,\, \nabla L_{\mathrm{frame}}\rangle$. By the explicit strong-convexity hypothesis, $\langle \delta\sigma,\, \nabla L_{\mathrm{frame}}\rangle \ge m_L\, |\delta\sigma|_g^2 = m_L\, V$, giving a control contribution $\le -2\eta\, m_L\, V$.

Drift term. The drift contribution is $-2\, g(\delta\sigma,\, \dot\sigma^*)$. By Cauchy–Schwarz: $|g(\delta\sigma,\, \dot\sigma^*)| \le |\delta\sigma|_g\, \|\dot\sigma^*\|_g = \sqrt{V}\, \|\dot\sigma^*\|_g$. Hence the drift contribution is bounded by $+2\sqrt{V}\, \|\dot\sigma^*\|_g$.

Combined. Adding both contributions gives (22). Part (a) follows by setting $dV/dt < 0$; part (b) by solving $dV/dt = 0$ for the fixed point $\sqrt{V_\infty} = \|\dot\sigma^*\|_g/(\eta m_L)$; part (c) by setting $\dot\sigma^* = 0$. □

5.3 Convergence Rate under Persistent Excitation

Definition 24 (Persistent excitation constant)**.** *The* persistent excitation constant $\alpha_{\mathrm{PE}} > 0$ *is the minimum eigenvalue of the time-averaged Fisher information matrix:*

$$\bar{g}(t) := \frac{1}{T}\int_t^{t+T} g(\sigma(s))\,ds \;\succeq\; \alpha_{\mathrm{PE}}\, I \qquad \text{for all } t, \tag{24}$$

when the residual likelihood is persistently exciting. Pointwise full-rank regularity and a positive environmental entropy rate do not by themselves provide a uniform lower bound.

Remark 25 (Tracking vs convergence)**.** *In the static case ($\sigma^* = \mathrm{const}$), Theorem 23(c) gives pure exponential convergence: $V(t) \le V(0)\, e^{-2\eta m_L t}$. Under environmental drift, convergence to zero is* not *possible—instead the loop maintains the mismatch within the tracking neighbourhood* (23)*. The tracking error V_∞ grows with drift speed $\|\dot{\sigma}^*\|_g$ and decreases with loop parameters η and m_L. The distinct constant α_{PE} controls statistical conditioning of the Fisher metric; no equality $m_L = \alpha_{\mathrm{PE}}$ is assumed. A physical implementation must separately relate adaptation rate to its energy budget.*

Remark 26 (The necessity of novelty)**.** *A frozen or uninformative residual likelihood can violate persistent excitation, but $h_\mu \to 0$ alone does not determine α_{PE} or m_L. Loss of excitation therefore has to be checked in the supplied observation model rather than inferred from entropy rate alone.*

5.4 Four-Part Model Template

The following is an architecture template for the declared model, not a necessity or sufficiency theorem for persistent agents.

Definition 27 (Four-part model architecture)**.** *The running T-DOME model uses four supplied structural layers:*

(I) ***External observable geometry.*** *The environmental observable algebra supports a metric structure; $Cl(1,3)$ serves as the running example throughout the programme, but the argument applies to any algebra satisfying (C1). This is a modelling premise; revised HAFF/Q-RAIF do not derive a general metric dictionary.*

(II) ***Internal control algebra.*** *The agent carries a supplied internal algebra $Cl(V,q)$ and a supplied representation map ϕ into an external algebra. Revised Q-RAIF neither uniquely selects the Clifford algebra nor proves a signature-preserving embedding.*

(III) ***Self-monitoring function.*** *The agent maintains a Lyapunov function $V(\sigma)$* (21) *satisfying the tracking bound* (22)*, via a second-order control loop on the agent's Fisher information, keeping the mismatch within the tracking neighbourhood* (23)*.* Conditional source: *Theorem 23, under its explicit local strong-convexity assumptions.*

(IV) ***Biased, non-Markovian memory.*** *The agent carries path-dependent state (non-Markovian memory kernel $\mathcal{K}(t,s)$) compressed through a gauge-fixed reference frame (the ego $\mathfrak{E}$). This is a model choice motivated by Papers I and II, not a necessity result inherited from them.*

The template is sufficient only for performance separately proved by a fully specified implementation. Omitting a layer may be benign in a different architecture.

6 Thermodynamic Cost

6.1 The Three Cost Components

One implementation may use sensing, computing, and actuation. Their thermodynamic costs depend on how they are physically realized.

1. Sensing cost. The meta-observer must read the prediction residuals from the ego's processing pipeline. Measurement or copying can in principle be reversible. If sensor registers irreversibly erase Shannon information at rate R_s bits per unit time, Landauer's principle gives

$$\dot{W}_{\text{sense}} \;\geq\; k_B T \,\ln 2 \cdot R_s. \tag{25}$$

No positive sensing bound follows merely from reading a stream.

2. Computing cost. Reversible logic likewise evades a gate-count Landauer charge. If computational work registers irreversibly erase Shannon information at rate R_c bits per unit time, then

$$\dot{W}_{\text{compute}} \;\geq\; k_B T \,\ln 2 \cdot R_c. \tag{26}$$

3. Actuating cost. Rotating the gauge parameter from the current frame σ to the estimated optimal frame $\hat{\sigma}^*$ is a finite-time thermodynamic transformation on the gauge manifold. By the Sivak–Crooks bound (Proposition 7):

$$\dot{W}_{\text{actuate}} \;\geq\; \frac{\mathcal{L}^2(\sigma,\, \hat{\sigma}^*)}{\tau^2_{\text{recalib}}}, \tag{27}$$

where $\mathcal{L}(\sigma,\, \hat{\sigma}^*)$ is the thermodynamic length (7) of the geodesic from σ to $\hat{\sigma}^*$, and τ_{recalib} is the recalibration time.

6.2 Conditional Implementation Bound

Proposition 28 (Conditional implementation cost)**.** *Suppose that (i) sensing, computing, and actuation use separable physical degrees of freedom with no shared erasure accounting; (ii) sensor and computational registers erase Shannon information at rates R_s and R_c; and (iii) frame actuation follows a linear-response protocol of duration τ_{recalib} and thermodynamic length $\mathcal{L}$. Then*

$$\dot{W}_{\text{loop}} \;\geq\; k_B T \,\ln 2 \,[R_s + R_c] + \frac{\mathcal{L}^2(\sigma,\, \sigma^*)}{\tau^2_{\text{recalib}}}. \tag{28}$$

This is a bound for the declared implementation, not a universal cost of sensing, Fisher-information evaluation, or self-reference.

Proof. Landauer's bound applies to the two declared irreversible reset streams, giving the first two terms. The Sivak–Crooks work bound divided by the protocol duration gives the actuation rate. The separability hypothesis permits addition without double counting. Dropping that hypothesis, or implementing reversible sensing/computation with no reset in the window, invalidates the corresponding additive terms. □

6.3 Withdrawal of the Universal Persistence Budget

Remark 29 (Withdrawal of the universal persistence budget)**.** *The former additive persistence corollary is withdrawn. Revised Papers I and II provide no universal memory-maintenance or ego-processing Landauer terms that can be inherited here. Proposition 28 may be combined with independently established costs for a concrete hardware implementation, provided separability and reset accounting are verified afresh.*

7 Worked Example: Qubit in a Drifting Two-Channel Bath

7.1 Model Setup

We extend the two-channel qubit model from Paper II (Section 6) by introducing environmental drift.

Inherited setup. A qubit ($\dim \mathcal{H}_S = 2$) with internal algebra $Cl(0,2) \cong \mathbb{H}$ ($D = 4$), coupled to two bosonic channels:

- Dephasing channel (σ_z): $J_z(\omega) = 2\lambda_z\gamma_z\omega/(\omega^2 + \gamma_z^2)$.
- Dissipative channel (σ_x): $J_x(\omega) = 2\lambda_x\gamma_x\omega/(\omega^2 + \gamma_x^2)$.

Paper II's ego selects $V_{\text{fg}} = \text{span}\{1, \mathbf{k}\}$ (the dephasing subspace), discarding $V_{\text{bg}} = \text{span}\{\mathbf{i}, \mathbf{j}\}$.

Environmental drift and residual channel. We use Paper II's revised linear mismatch model and the Gaussian residual channel of Proposition 14:

$$\theta(t) = \varepsilon t, \qquad e_m \mid \theta_m \sim \mathcal{N}(a\theta_m^2, \sigma_e^2), \quad a = 1.2, \quad \sigma_e = 0.35, \quad \varepsilon = 0.04\,\omega_0. \tag{29}$$

At sampling rate $r = 50\,\omega_0$, the visibility crossover is $t_{\text{vis}} = \pi/(4\varepsilon) \approx 19.63\,\omega_0^{-1}$. The Gaussian likelihood is a declared observation model, not derived from the bosonic spectral density.

Parameter mapping.

Quantity	Value	Source
$D = \dim Cl(0,2)$	4	Paper II
k^*	2	chosen Paper II model frame
$\mathcal{C}_{\text{budget}}$	$2\,h_\mu$	Paper II
θ_0 (initial misalignment)	0	fresh calibration
ε (angular drift rate)	$0.04\,\omega_0$	Eq. (29)
t_{vis}	$19.63\,\omega_0^{-1}$	Paper II visibility model
η (adaptation rate)	0.5	meta-observer

7.2 Fisher Information under Drift

For the declared residual channel, the per-sample and accumulated local Fisher signals are

$$\mathcal{I}_1(t) = \frac{4a^2\varepsilon^2 t^2}{\sigma_e^2}, \qquad J_T = \frac{4a^2 r\varepsilon^2 T^3}{3\sigma_e^2}. \tag{30}$$

The first quantity grows quadratically in time; the second grows cubically because it also counts samples. For $J_{\min} = 20$, Equation (15) gives $T_{\text{det}} \approx 2.52\,\omega_0^{-1} < t_{\text{vis}}$. This ordering is a parameter check, not a general theorem.

7.3 Loop Dynamics: Self-Calibration in Action

Under the natural gradient update (19), the frame angle $\phi(t)$ tracks the drifting optimal frame $\phi^*(t)$. The Lyapunov function $V(t) = (\phi(t) - \phi^*(t))^2$ is governed by the tracking bound (22): the loop drives V toward the tracking neighbourhood $V_\infty = \|\dot{\sigma}^*\|_g^2/(\eta m_L)^2$, with the approach rate set by the loss strong-convexity modulus m_L and adaptation rate η.

Comparison.

- **Without loop** (Paper II agent): the mismatch grows as $\theta(t) = \varepsilon t$, reaching the visibility crossover at t_{vis}.
- **With loop** (Paper III agent): the mismatch oscillates around zero, bounded by the estimation neighbourhood supplied by Theorem 23 when its local strong-convexity and bounded-drift hypotheses hold.

A multi-dimensional numerical evaluation extending this qubit illustration to continuous drift is presented in Section 8.

7.4 Thermodynamic Cost Evaluation

For illustration, suppose a particular implementation irreversibly resets sensor and work registers at rates $R_s = 2$ and $R_c = 1$ bits per unit time and uses separable actuation

hardware. Then

$$\dot{W}_{\text{sense}} \geq k_B T \ln 2 \cdot R_s = 2\, k_B T \ln 2, \tag{31}$$

$$\dot{W}_{\text{compute}} \geq k_B T \ln 2 \cdot R_c = k_B T \ln 2, \tag{32}$$

$$\dot{W}_{\text{actuate}} \geq \frac{\mathcal{L}^2}{\tau_{\text{recalib}}^2} \approx \frac{(\Delta\theta)^2\, \tau_{\text{relax}}}{\tau_{\text{recalib}}^2}\, k_B T. \tag{33}$$

The two Landauer terms are present because this example stipulates irreversible reset. A reversible implementation may lower them, and no percentage overhead relative to Papers I or II follows.

8 Numerical Demonstration

We report two separate numerical checks. The first directly samples the Gaussian likelihood and computes its score-based Fisher signal. The second retains the earlier multi-dimensional calibration controller, whose top-k gradient-overlap sentinel is a heuristic proxy rather than Fisher information.

8.1 Direct Fisher-Signal Test

We sample Equation (12) at rate $r = 50$ with $a = 1.2$, $\sigma_e = 0.35$, $\varepsilon = 0.04$, and 4,000 Monte Carlo residuals per plotted time. For each residual we evaluate the score

$$s_\theta(e) = \frac{(e - a\theta^2)(2a\theta)}{\sigma_e^2}$$

and estimate $\mathcal{I}_1(\theta) = \mathbb{E}[s_\theta^2]$ directly. Figure 1 shows agreement with the analytic quadratic curve and with the cubic accumulated curve. For $J_{\min} = 20$, the analytic and empirical crossings are respectively $T_{\text{det}} = 2.517$ and 2.524, both before $t_{\text{vis}} = 19.635$. This verifies Proposition 14 and Corollary 16 for the declared parameter set; it does not establish the ordering for other likelihoods or thresholds.

8.2 Heuristic Calibration-Control Model

Environment. A d-dimensional linear prediction task: $y(t) = \mathbf{w}(t)^\top \mathbf{x}(t) + \sigma\, \epsilon(t)$, $\mathbf{x}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, $\epsilon \sim \mathcal{N}(0, 1)$. The weight vector $\mathbf{w}(t) \in \mathbb{S}^{d-1}$ drifts by receiving random perturbations on *background* dimensions only (indices $k, \ldots, d-1$), then renormalising. Signal therefore migrates progressively from the ego's foreground to its blind sector.

Agents.

- **Fixed ego** (Paper II analogue): learns a linear model on a *fixed* foreground subspace of dimension k via stochastic gradient descent (SGD, rate η, decay λ). Embodies the "frozen gauge" of Paper II.

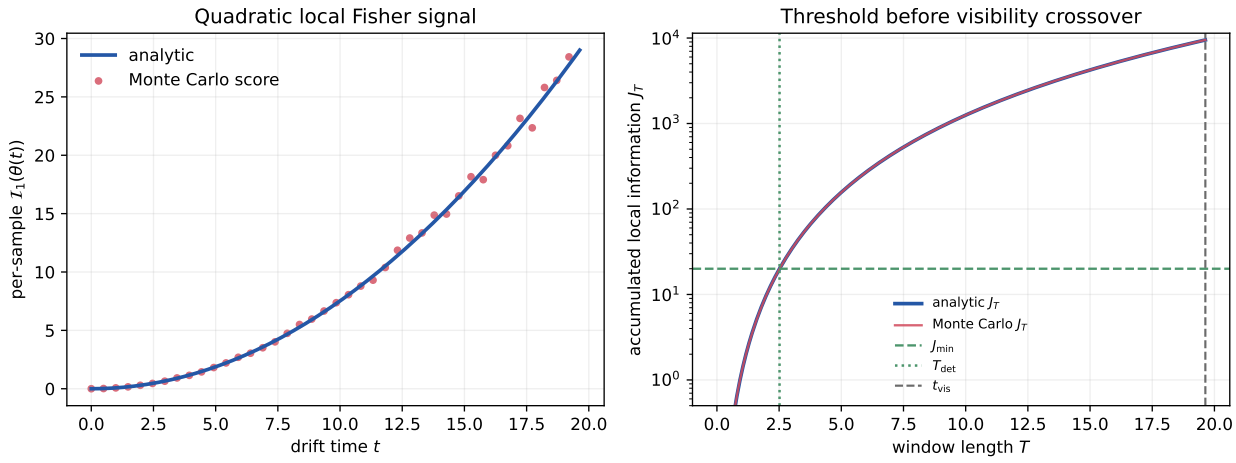


Figure 1: **Direct score-based Fisher test. Left:** analytic per-sample Fisher information and Monte Carlo score-squared estimate, both quadratic in drift time. **Right:** analytic and empirical accumulated local information; the $J_{\min} = 20$ crossing precedes the independently defined visibility crossover for this parameter set.

- **Calibrated loop** (Paper III analogue): identical ego plus a staleness sentinel and recalibration mechanism. The sentinel tracks $g_i = \text{EMA}(|e\, x_i|)$ for each dimension i (an absolute-gradient proxy), computes the fraction of top-k gradient dimensions *not* in the current foreground as a frame-staleness index $m \in [0, 1]$, and triggers recalibration when $\text{EMA}(m) > \theta$. After recalibration, a settling period of τ steps elapses before the sentinel resumes monitoring.

Parameters.

Quantity	Value	Role
d	20	full ambient dimension
k	5	ego foreground dimension ($k/d = 0.25$)
σ	0.1	observation noise std
η	0.01	SGD learning rate
λ	0.998	SGD weight decay
θ	0.25	staleness threshold
Λ_{demo}	variable	simulator drift amplitude per step
τ	variable	settling period (cooldown)

Oracle metrics. Neither agent has access to $\mathbf{w}(t)$. We evaluate performance externally using the *oracle full-space error*:

$$\mathcal{E}_{\text{full}} = \left\|\mathbf{w}_{\text{ego}} - \mathbf{w}^*_{\text{fg}}\right\|^2 + \left\|\mathbf{w}^*_{\text{bg}}\right\|^2, \tag{34}$$

where $\mathbf{w}^*_{\text{fg}}$ and $\mathbf{w}^*_{\text{bg}}$ denote the true weight vector restricted to foreground and background coordinates respectively, and $\mathbf{w}_{\text{ego}}$ is the ego's foreground-supported estimator lifted to the full space. The first term captures foreground tracking error (accessible to the ego); the second captures hidden-sector signal (invisible).

8.3 Results

Result 1: Delusion-correction separation (Figure 2). At simulator drift amplitude $\Lambda_{\text{demo}} = 0.02$, settling period $\tau = 200$, and $T = 5\,000$ steps, three phenomena are visible:

(a) *Delusion trap.* The ego's foreground tracking error converges to near zero, while the true full-space error rises toward ~ 1 and stabilises. The growing gap between the two is the hidden sector in this supplied linear model. It does not prove a no-go theorem for all first-order controllers.

(b) *Detectability.* The staleness sentinel produces a clean sawtooth: rising from zero after each recalibration, crossing the threshold $\theta = 0.25$, and triggering frame reset (25 events over $T = 5\,000$). This demonstrates that the chosen overlap proxy can trigger a useful reset. The sentinel is not Fisher information; the direct Fisher test is reported in Section 8.1.

(c) *Net benefit.* The calibrated loop achieves $\mathcal{E}_{\text{full}} \approx 0.74$ versus the fixed ego's ≈ 1.02: a 27% reduction in true prediction error.

Result 2: Phase structure and optimal calibration budget (Figures 3–4). We scan 16 simulator drift amplitudes $\Lambda_{\text{demo}} \in [0.005, 0.08]$ and 16 settling periods $\tau \in [15, 800]$ (logarithmically spaced), running both agents for $T = 4\,000$ steps across 6 random seeds per grid point.

Figure 3(a) shows the performance gain $\Delta = \mathcal{E}_{\text{ego}} - \mathcal{E}_{\text{loop}}$: the loop improves over the ego (green) across most of the parameter space, with a boundary at $\Delta = 0$ (dashed) below which recalibration is counterproductive (very low drift, where the overhead of re-learning exceeds the benefit of tracking). The solid curve traces the *optimal settling period* $\tau_{\text{opt}}(\Lambda_{\text{demo}})$—the recalibration period minimising $\mathcal{E}_{\text{loop}}$—which decreases monotonically from ~ 370 steps at $\Lambda_{\text{demo}} = 0.005$ to ~ 100 at $\Lambda_{\text{demo}} = 0.08$. Figure 3(b) shows that calibration frequency increases smoothly with drift and with shorter settling period, exhibiting the cost–performance trade-off of this supplied controller and cost proxy. It does not test Proposition 28.

Extracting $\tau_{\text{opt}}(\Lambda_{\text{demo}})$ yields the *optimal calibration frequency* $\alpha_{\text{opt}}(\Lambda_{\text{demo}}) = 1/\tau_{\text{opt}}$ (Figure 4). The curve is smooth and monotonically increasing: faster drift demands tighter calibration. It saturates at high Λ_{demo} near $\alpha_{\text{opt}} \approx 0.01$ per step ($\tau_{\text{opt}} \approx 100$), of the same order as the learner's settling time. This is consistent with the intuition that drift estimation requires a minimum observation window; the self-referential Cramér–Rao bound (Theorem 18) provides the analytic counterpart of this computational floor.

8.4 Scope of This Demonstration

This demonstration **does** show:

1. The explicit Gaussian residual model has a directly measured score-squared Fisher signal with the quadratic and cubic scalings in Proposition 14.

2. For the declared Gaussian parameters, the Fisher threshold is crossed before the linear visibility crossover.

3. A separate, non-Fisher frame-staleness proxy has clean threshold dynamics and triggers effective recalibration in the supplied linear controller.

4. An optimal calibration frequency $\alpha_{\text{opt}}(\Lambda_{\text{demo}})$ exists, increases monotonically with drift rate, and saturates at the learner's settling timescale in the scanned grid.

In summary, the direct experiment validates the declared Fisher likelihood, while the controller experiment establishes a model-specific proxy trade-off. Neither is universal across architectures or environment classes.

This demonstration does **not** show:

1. That the specific functional form of $\alpha_{\text{opt}}(\Lambda)$ matches the analytic Cramér–Rao prediction in the large-d limit. The demonstration confirms the monotonic trend and saturation; deriving the exact scaling exponent from Theorem 18 remains open.

2. That the results generalise to all environment classes. The model uses Gaussian features, linear regression, and isotropic background drift; extensions to non-linear, non-Gaussian, or structured-drift settings require further investigation.

3. That the calibration loop is optimal among all possible adaptive strategies. It implements one specific realisation of the calibration-loop architecture.

4. That the gradient-overlap sentinel estimates Fisher information, or that recalibration-event frequency is a thermodynamic dissipation measurement.

Reproducibility. The controller simulation is in `tdome_demo.py`. The direct Fisher experiment is in `fisher_residual_demo.py`, with raw output in `fisher_residual_data.csv`. Both require only NumPy and Matplotlib and use fixed random seeds.

9 Discussion

9.1 Summary of Results

Result	Statement	Sec.
Even-likelihood non-identifiability	The Gaussian residual record cannot identify mismatch sign	3.1
Drift Detectability	Per-sample FI is quadratic and accumulated local FI is cubic in the Gaussian linear-drift model	3.3
Self-Referential CR Bound	Drift estimation bounded by $1/(n_{\text{eff}}\,\mathbb{E}_\pi[\mathcal{I}_1] + \mathcal{I}_{\text{ego}})$ under regular prior/likelihood data	4.2
Loop Tracking Bound	Lyapunov V with tracking neighbourhood $V_\infty = \|\dot{\sigma}^*\|^2/(\eta m_L)^2$ under local geodesic strong convexity	5.2
Four-Part Structure	Four supplied layers define the running model architecture	5.4
Loop Cost	$\dot{W}_{\text{loop}} \geq k_B T \ln 2\,(R_s + R_c) + \mathcal{L}^2/\tau^2_{\text{recalib}}$ for a separable reset-based implementation	6.2
Persistence Budget	Universal inherited budget withdrawn	6.3
Numerical Demonstration	Direct Fisher test plus a separate heuristic-controller scan	8

9.2 Conditional Interface Sequence

Papers I–III provide a conditional interface sequence:

Paper	Crisis	Resolution	What is born
Paper I	Restricted autonomous baseline	Correlation-aware accounting	**Conditional resource model**
Paper II	Raw archive exceeds a declared budget	Supplied coordinate projection	**Frame model**
Paper III	Linear visibility loss in a fixed frame	Likelihood-based detection; conditional tracking	**Calibration model**

The arrows are motivation or model interfaces. The tracking theorem is local and conditional, the universal persistence budget is withdrawn, and no termination or completeness result follows.

9.3 What This Paper Does and Does Not Show

This paper **does** show:

1. The even Gaussian residual likelihood cannot identify mismatch sign, while its score has the explicit Fisher scalings in Proposition 14.
2. Detection precedes visibility loss exactly when the threshold condition in Corollary 16 holds.
3. Drift estimation precision is bounded by the Self-Referential Cramér–Rao bound (Theorem 18).
4. The calibration loop tracks the optimal frame within a bounded neighbourhood under a Lyapunov tracking bound (Theorem 23).
5. A separable reset-based hardware model obeys the conditional cost bound in Proposition 28.

This paper does **not** show:

1. That self-referential calibration implies or requires phenomenal consciousness, subjective experience, or qualia. "Reflexivity" as used here denotes second-order control, nothing more.
2. That the Lyapunov function V is a measure of "awareness." It is a control-theoretic stability condition, not a consciousness metric.
3. That the four-part template is necessary, sufficient, or a complete characterisation of agency.
4. That Fisher information requires the agent to "know" it is computing Fisher information. The computation can be implemented implicitly by any physical system whose dynamics approximate the natural gradient.
5. That the calibration loop eliminates the ego's bias. It tracks and compensates for drift in the bias; the four bias terms of Paper II persist.
6. That sensing or computing is intrinsically dissipative. The Landauer terms require irreversible register reset.
7. That this framework applies to all possible systems. It applies to systems satisfying (C1)–(C5).
8. That the structural parallel with philosophical concepts of self-awareness constitutes a philosophical or metaphysical claim.
9. That the Clifford algebra is the only possible algebraic setting. Other control algebras may yield analogous results with different quantitative bounds.

We have established a checkable Gaussian Fisher signal, a regular Bayesian estimation bound, and a local tracking result. Their use in a physical agent remains conditional on the declared observation, optimization, and hardware models.

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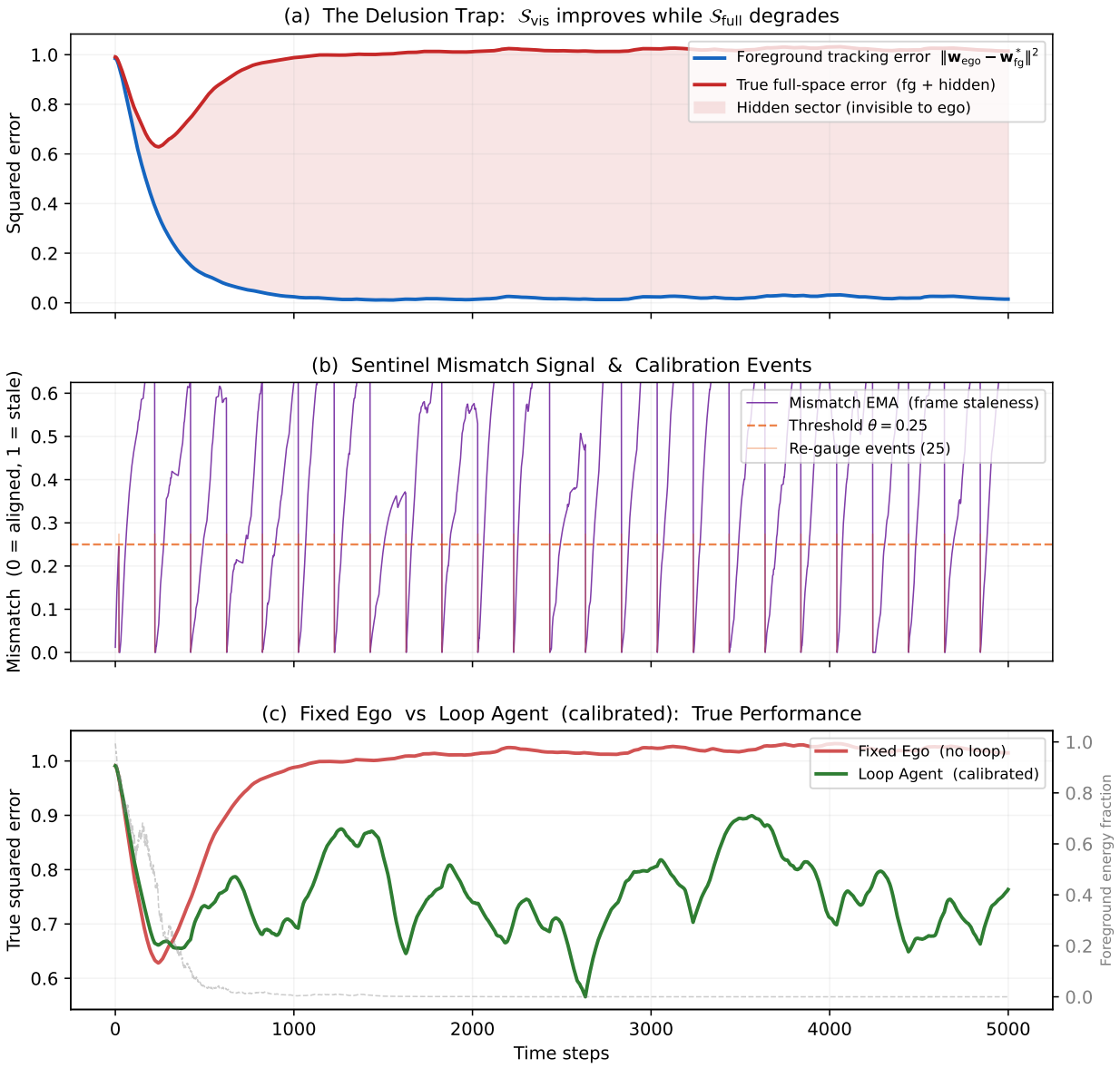


Figure 2: **Delusion trap and calibration loop.** $d = 20$, $k = 5$, $\Lambda_{\text{demo}} = 0.02$, $\tau = 200$, $T = 5\,000$. **(a)** Foreground tracking error (blue) decreases toward zero while true full-space error (red) increases; the shaded region is the hidden sector, invisible to the ego. **(b)** Frame-staleness sentinel (purple) rises monotonically between recalibration events (orange), producing a sawtooth with 25 threshold crossings. **(c)** The calibrated loop (green) maintains lower true error than the fixed ego (red); the grey dashed line shows the foreground energy fraction decaying as signal migrates to the background.

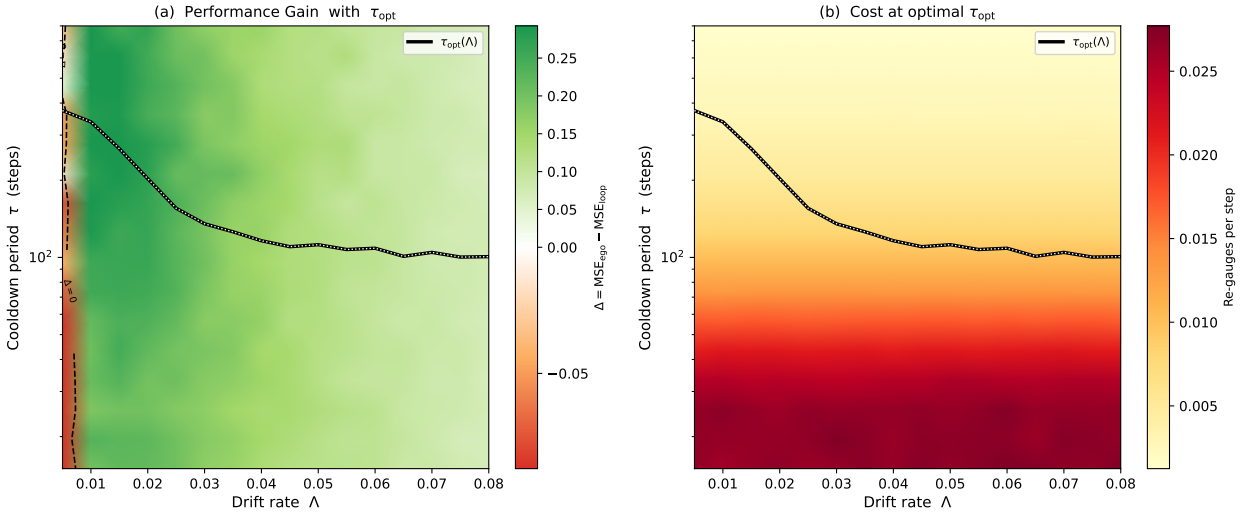


Figure 3: **Phase structure and calibration cost.** 16×16 grid, $T = 4\,000$, 6 seeds per point. **(a)** Performance gain Δ; green = loop improves on ego, red = counterproductive. Solid curve: $\tau_{\text{opt}}(\Lambda_{\text{demo}})$. Dashed: $\Delta = 0$ boundary. **(b)** Calibration frequency (thermodynamic cost proxy: recalibration events per step, proportional to energy expenditure under a fixed per-recalibration cost model); τ_{opt} overlaid.

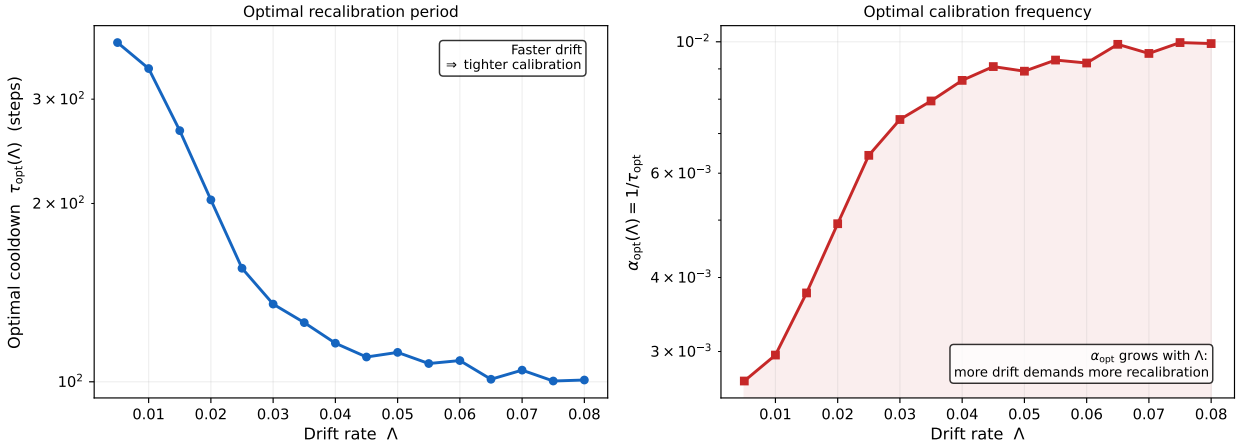


Figure 4: **Optimal calibration frequency. Left:** $\tau_{\text{opt}}(\Lambda_{\text{demo}})$ decreases monotonically with drift rate. **Right:** $\alpha_{\text{opt}}(\Lambda_{\text{demo}}) = 1/\tau_{\text{opt}}$ increases with drift rate and saturates at the learner's settling timescale ($\sim$100 steps), consistent with an observation-window floor.